

Nate Haris

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EDUCATION

Cornell University

Graduation June 2028

Department of Electrical Engineering | GPA: 3.5

Ithaca, NY

- *Relevant Coursework:* Circuit Analysis, Computer Architecture, Embedded OS Design, Computer Systems Programming, Microelectronics Design

Lexington High School

2020 – 2024

GPA: 4.01

Lexington, MA

PROJECTS

Projects are covered in depth in my personal portfolio: nh428.github.io

Custom Dashboard PCB

- Developed, brought up, and tested custom dashboard PCB and driver modules for Cornell's race car, learning about PCB design principals.

Pipelined Multi-Core Processor

- Designed and coded a pipelined 4 core processor to execute a subset of RISC-V assembly commands, optimized for sorting algorithms.
- Coded in synthesizable Verilog with optimization techniques such as branch target buffers, parallel cache access, ring network memory, and forwarding.

Cybersecurity Research

- Conducted Cybersecurity research and published to Cornell's ArXiv database.
- Developed a programmatic open-source tool to assess a mobile application for significant API endpoint vulnerabilities.

EXPERIENCE

Senior Audio and Visual Technician – Cornell A&S AV

Nov. 2024 – Present

Ithaca, NY

- Responsible for over 200 integrated AV systems across campus, maintaining visual, audio, and network functionality.
- Implemented and tested new technologies for client satisfaction, including touch panel and seamless video streaming capabilities.
- Assisted with onboarding process of 5+ new employees, developed flowcharts to aide with common problems.

Student Manager – Cornell Engineering Learning Lab

Oct. 2024 – Present

Ithaca, NY

- Overseen operations of lab space for 36 project teams of 1,800+ students with \$100,000s of annual funding.
- Taught safe operation practices of heavy machinery, including CNC, lathes, milling machines, various types of saws, and other power tools. item Learned various general mechanical and fabrication skills, such as carbon fiber preparation, welding, and how to work with high voltage systems.

ACTIVITIES

- Volunteer Firefighter at local department, Cayuga Heights
- Cornell Club-A Team Ultimate Frisbee Player
- CKB Residence hall Vice-President
- Cybersecurity Club member
- Preorientation backpacking program, Outdoor Odyssey, Guide and Selections committee member
- Amateur vehicle mechanist; Amateur winemaker

SKILLS

Firmware Skills: Verilog, Assembly, Java, C, C++, Python, Html, MATLAB

Technical Skills: PCB design in Altium, Circuit Analysis with LTSPICE, AutoCAD, Software development, FPGA development, Cybersecurity